



# Koneru Lakshmaiah Education Foundation

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## DEPARTMENT OF CSE

### PROJECT ABSTRACT SUBMISSION

COURSE: COMPILER DESIGN (24CS2235F)

A.Y. 2026 - 2027

| TITLE OF THE PROJECT | Log Query Optimization System |                     |
|----------------------|-------------------------------|---------------------|
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## Abstract

The Log Query Optimization System is a compiler-based application designed to analyze, validate, and optimize queries used for searching and processing log data. Log files contain important information about system activities, errors, warnings, and user actions. As the volume of log data increases, inefficient queries can require more processing time and resources. The proposed system addresses this issue by applying fundamental Compiler Design techniques to log query processing.

The system processes user queries through various compiler phases, including lexical analysis, syntax analysis, semantic analysis, and intermediate representation. Lexical analysis identifies keywords, identifiers, operators, and values, while syntax and semantic analysis verify the correctness and validity of the query. The validated query is then converted into an intermediate representation for further processing.

The query optimization phase improves the query by removing redundant conditions, simplifying logical expressions, and arranging filtering operations efficiently. The optimized query is then generated for execution, reducing unnecessary processing and improving efficiency. Overall, the project demonstrates the practical application of compiler design principles in log analysis and provides a structured approach for faster and more effective log query processing.

Signature of the Guide

HOD